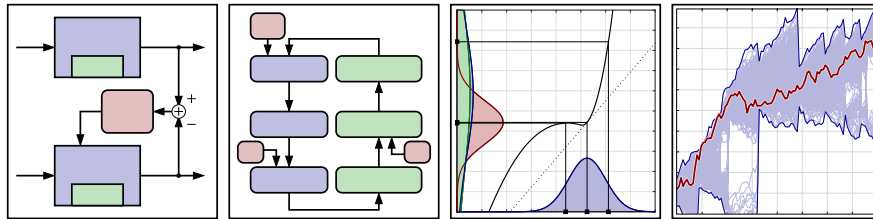


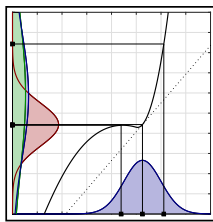


Welcome to the course!

- Welcome to **Nonlinear Kalman Filters!**
- This course is the third in a specialization that investigates the design and application of Kalman filters.
- This course focuses on generalization of the linear Kalman filter to problems requiring estimating the state of nonlinear systems.



What topics will we study in this course?

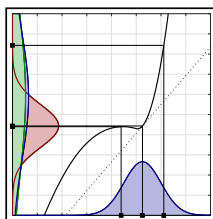


In this course, you will learn:

- How to derive the steps of the nonlinear extended Kalman filter (EKF).
- How to implement an EKF in Octave code.
- How to derive the steps of the nonlinear sigma-point Kalman filter (SPKF, a.k.a. UKF).
- How to implement an SPKF in Octave code.
- Some extensions and refinements, including improving the accuracy of EKF, simplifying the SPKF, and how to tune a KF adaptively.
- How to apply nonlinear KFs to estimate the parameters of a dynamic model.



What skills will you gain in this course?

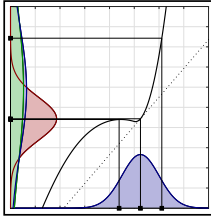


After completing the course, you'll be able to:

- Execute provided Octave code that implements the EKF and interpret its results.
- Execute provided Octave code that implements the SPKF and interpret its results.
- Execute provided Octave code that implements adaptive nonlinear KFs, cubature Kalman filters, and iterated extended Kalman filters.
- Perform parameter estimation using nonlinear KFs.



Prerequisites; for further study



Prerequisites:

■ Course 1: ***Kalman-filter boot camp.***

- Basic background in state-space models, random variables and processes, state estimation with Kalman filters.

■ Course 2: ***Linear Kalman filter deep dive.***

- How to derive the linear Kalman filter, how to adjust the filter steps when the derivation assumptions are violated.

Optional resource:

- There is no required textbook for this course.
- But there are some good texts on KF. One that I like and use in my UCCS graduate course “Applied Kalman Filtering” is:
 - Dan Simon, *Optimal State Estimation: Kalman, H_∞ , and Nonlinear Approaches*, Wiley Interscience, New Jersey, 2006.